

COMP4436 AIoT Project - Code Running Instructions

HOW TO RUN THE CODE:

1. Open Anaconda Prompt
2. Install required packages:
`pip install torch torchvision numpy matplotlib pandas jupyter`
3. Navigate to the folder containing AIoT_Project.ipynb
4. Launch Jupyter Notebook:
`jupyter notebook`
5. Open AIoT_Project.ipynb and run all cells (Kernel -> Restart and Run All)

MAPPING BETWEEN CODE OUTPUT AND REPORT:

Cell 20 - Model comparison table (parameters, size, MACs, inference time)
Maps to Section 3 (Implementation) - Model characteristics table

Cell 21 - Combined experiment results table (4 scenarios)
Maps to Section 4 (Experiments) - Main results table

Cell 21 - Side-by-side comparison plots (queue, Aol, latency)
Maps to Figures 2, 3, 4 in Section 4 (Experiments)

Cell 13 - Overload test with queue buildup and latency increase
Maps to overload demonstration in Section 4 (Experiments)

Cell 16 - Edge vs cloud inference comparison
Maps to Section 5 (Discussion) - Edge-cloud trade-off

Cell 19 - Model pruning demonstration
Maps to Section 3 (Implementation) - Model compression

NOTES:

- Total runtime: approximately 15-20 minutes
- MNIST downloads automatically on first run
- Dropout warnings can be ignored